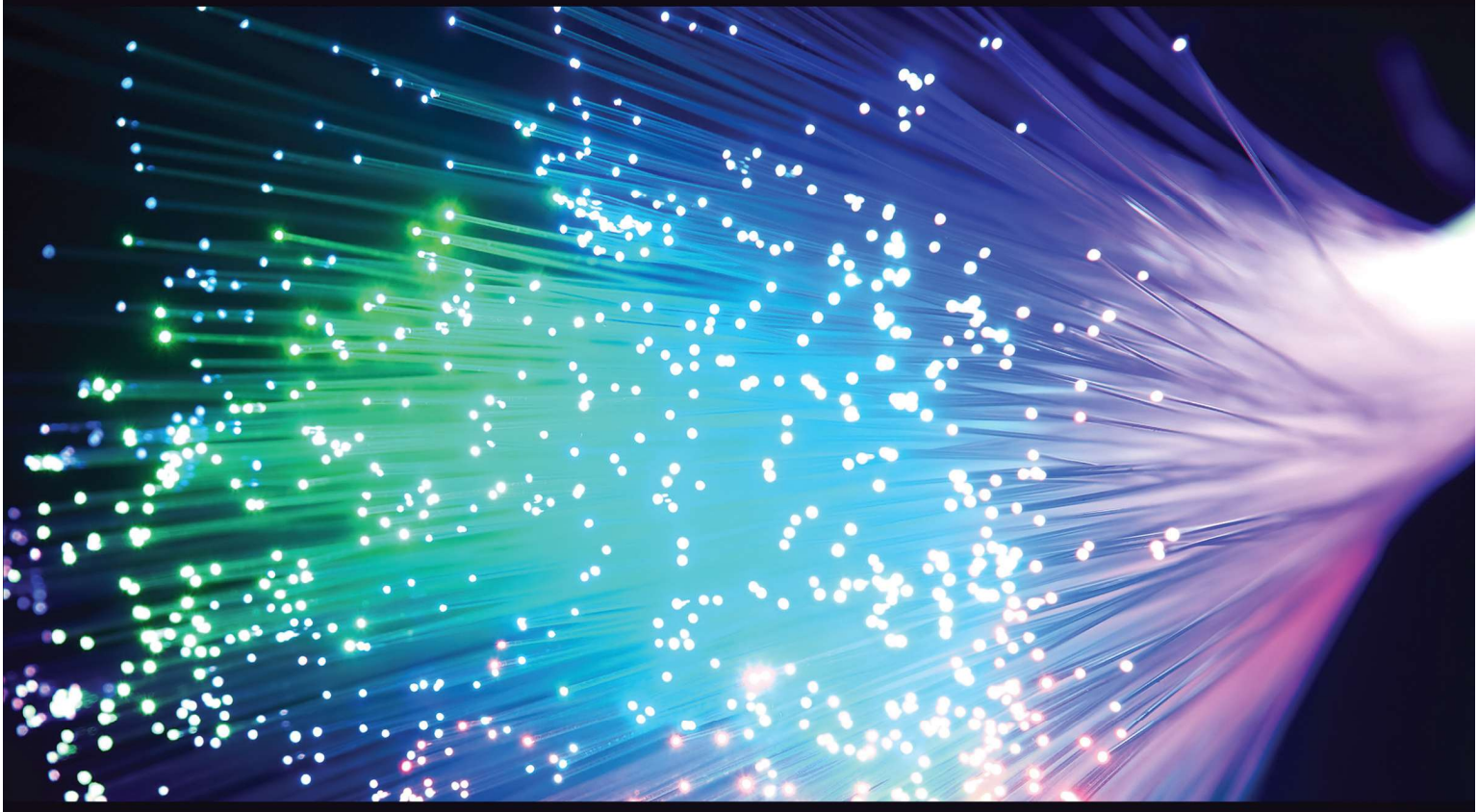
Eighth Edition

Systems Analysis and Design

Alan Dennis • Barbara Haley Wixom
Roberta M. Roth



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EIGHTH EDITION

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To Kelly

To Chris, Haley, and Hannah

To my dear friends, Sue and Karen. Thanks for all your care and support.

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Preface

Purpose of This Book

Systems Analysis and Design (SAD) is an exciting, active field in which analysts continually learn new techniques and approaches to develop systems more effectively and efficiently. However, there is a core set of skills that all analysts need to know no matter what approach or methodology is used. All information systems projects move through the four phases of planning, analysis, design, and implementation; all projects require analysts to gather requirements, model the business needs, and create blueprints for how the system should be built; and all projects require an understanding of organizational behavior concepts like change management and team building.

This book captures the dynamic aspects of the field by keeping students focused on doing SAD while presenting the core set of skills that we feel every systems analyst needs to know today and in the future. This book builds on our professional experience as systems analysts and on our experience in teaching SAD in the classroom.

This book will be of particular interest to instructors who have students do a major project as part of their course. Each chapter describes one part of the process, provides clear explanations on how to do it, gives a detailed example, and then has exercises for the students to practice. In this way, students can leave the course with experience that will form a rich foundation for further work as a systems analyst.

Outstanding Features

A Focus on Doing SAD The goal of this book is to enable students to do SAD—not just read about it, but understand the issues so that they can actually analyze and design systems. The book introduces each major technique, explains what it is, explains how to do it, presents an example, and provides opportunities for students to practice before they do it in a real-world project. After reading each chapter, the student will be able to perform that step in the system development life cycle (SDLC) process.

Rich Examples of Success and Failure The book includes a running case about a fictitious company called DrōnTeq. Each chapter shows how the concepts are applied in situations at DrōnTeq. Unlike running cases in other books, this text focuses examples on planning, managing, and executing the activities described in the chapter, rather than on detailed dialogue between fictitious actors. In this way, the running case serves as a template that students can apply to their own work. Each chapter also includes numerous Concepts in Action boxes that describe how real companies succeeded—and failed—in performing the activities in the chapter. Many of these examples are drawn from our own experiences as systems analysts.

Incorporation of Object-Oriented Concepts and Techniques The field is moving toward object-oriented concepts and techniques, often by gradually incorporating object-oriented concepts into traditional techniques. We have integrated several object-oriented concepts into our discussion of traditional techniques, although this may not be noticed by the students because few concepts are explicitly labeled as object-oriented concepts. For example, we include the

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development of use cases as the first step in process modeling (i.e., data flow diagramming) in Chapter 4, and the use (and reuse) of standard interface templates and use scenarios for interface design in Chapter 8.

Real-World Focus The skills that students learn in a systems analysis and design course should mirror the work that they ultimately will do in real organizations. We have tried to make this book as "real" as possible by building extensively on our experience as professional systems analysts for organizations such as IBM, the U.S. Department of Defense, and the Australian Army. We have also worked with diverse industry advisory boards of IS professionals and consultants in developing the book and have incorporated their stories, feedback, and advice throughout. Many students who use this book will eventually apply the skills on the job in a business environment, and we believe that they will have a competitive edge by understanding what successful practitioners feel is relevant in the real world.

Project Approach We have presented the topics in this book in the SDLC order in which an analyst encounters them in a typical project. Although the presentation necessarily is linear (because students have to learn concepts in the way in which they build on each other), we emphasize the iterative, complex nature of SAD as the book unfolds. The presentation of the material should align well with courses that encourage students to work on projects because it presents topics as students need to apply them.

Graphic Organization The underlying metaphor for the book is doing SAD through a project. We have tried to emphasize this graphically throughout the book so that students can better understand how the major elements in the SDLC are related to each other. First, at the start of every major phase of the system development life cycle, we present a graphic illustration showing the major deliverables that will be developed and added to the electronic "project binder" during that phase. Second, at the start of each chapter, we present a checklist of key tasks or activities that will be performed to produce the deliverables associated with this chapter. These graphic elements—the deliverables tied to each phase and the task checklist tied to each chapter—can help students better understand how the tasks, deliverables, and phases are related to and flow from one to another.

Finally, we have highlighted important practical aspects throughout the book by marking boxes and illustrations with a "push pin." These topics are particularly important in the practical day-to-day life of systems analysts and are the kind of topics that junior analysts should pull out of the book and post on the bulletin board in their office to help them avoid costly mistakes.

What's New in the Eighth Edition

The eighth edition features several changes from the seventh edition. A new chapter (Chapter 13) has been added that introduces students to Agile development, focusing specifically on the Scrum approach. While this book will always emphasize structured systems development, the increasingly widespread use of Agile methods in practice makes it important that students today have a basic appreciation of the characteristics, concepts, processes, and roles associated with Agile development.

To accommodate the new material in the eighth edition, some project management-oriented material was eliminated. The coverage of use cases and data flow diagramming was merged into a single chapter that emphasizes the way these two tools reinforce the analyst's understanding of processes.

Throughout the book, the chapter objectives have been revised to reflect active learning objectives. Chapter references to outside sources have been updated to current resources wherever possible. New Concepts in Action features appear throughout the book to provide updated,

real-world illustrations of the textbook content. A number of new minicases have also been added. Many of the minicases include directions to the instructor on how to use the minicase in a flipped (or inverted) classroom. Flipping the Systems Analysis and Design class enables instructors to employ active learning activities while in the classroom with students, enhancing student engagement and understanding.

For this edition, a series of tutorial lessons have been created that will teach students how to use and apply the Visible Analyst™ computer-assisted software engineering (CASE) software to a simple systems development project scenario. (Instructors should visit the instructor's website at www.wiley.com/go/dennis/sad8e for information about these lessons.)

We know that most professors and students find the Systems Analysis and Design class to have a lot of demanding content, particularly in those classes that include a significant project. Many professors would like their students to be able to experience first-hand how useful a CASE tool is to a systems analyst, but find it difficult to include instruction on a CASE tool in an already full course. The goal of these lessons is to enable students to learn the basics of the Visible Analyst CASE software with little involvement on the part of the professor. The lesson material is found on the student website for this textbook at www.wiley.com/go/dennis/sad8e. Professors have the flexibility to assign these tutorial lessons if they want to include the Visible Analyst software in their courses, but are also free to exclude this material if they prefer. The tutorial lessons have been written to provide students with a sufficient foundation to apply Visible Analyst to a more significant systems development project, should that be a part of their course.

Organization of This Book

This book is organized by the phases of the systems development life cycle (SDLC). Each chapter has been written to teach students specific tasks that analysts need to accomplish over the course of a project, and the deliverables that will be produced from the tasks. As students complete the book, tasks will be "checked off" and deliverables will be completed and stored in project folders.

Part 1 covers the first phase of the SDLC, the Planning Phase. Chapter 1 introduces the SDLC, the roles and skills needed for a project team, project initiation, the systems request, and feasibility analysis. Chapter 2 discusses project selection, the selection of an SDLC methodology for the project, and project management, with emphasis on the staffing plan, project charter, risk assessment, and tools used to help manage and control the project.

Part 2 presents techniques needed during the analysis phase. In Chapter 3, students are introduced to requirements determination and are taught a variety of analysis techniques to help with business process automation, business process improvement, and business process reengineering. Chapter 4 focuses on use cases and process models, and Chapter 5 explains data models and normalization.

The Design Phase is covered in Part 3 of the textbook. In Chapter 6, students create an alternative matrix that compares custom, packaged, and outsourcing alternatives. Chapter 7 focuses on designing the system architecture, which includes the architecture design, hardware/software specification, and security plan. Chapter 8 focuses on the user interface and presents interface design; in this chapter, students learn how to create use scenarios, the interface structure diagram, interface standards, and interface prototypes. Finally, data storage design and program design are discussed in Chapters 9 and 10, which contain information regarding the data storage design, the program structure chart, and program specifications.

The Implementation Phase is presented in Chapters 11 and 12. Chapter 11 focuses on system construction, and students learn how to build and test the system. It includes information about the test plan and user documentation. Conversion is covered in Chapter 12, where students learn about the conversion plan, the change management plan, the support plan, and the project assessment.

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In this edition, Agile Development concepts are introduced in Chapter 13. Following an introduction to Agile values and principles, students get an in-depth look at the Scrum development method.

Supplements

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Instructors Manual The instructor's manual provides resources to support the instructor both in and out of the classroom:

- Short experiential exercises can be used to help students experience and understand key topics in each chapter.
- Short stories have been provided by people working in both corporate and consulting environments for instructors to insert into lectures to make concepts more colorful and real.
- Additional mini-cases for every chapter allow students to perform some of the key concepts that were learned in the chapter.
- Answers to end-of-chapter questions and exercises are provided.

Power Point Slides PowerPoint slides are provided that instructors can tailor to their classroom needs and that students can use to guide their reading and studying activities.

Test Bank Test Bank includes a variety of questions ranging from multiple choice to essay-style questions. A computerized version of the Test Bank is also available.

Wiley E-Text: Powered by VitalSource This Wiley e-text offers students continuing access to materials for their course. Your students can access content on a mobile device, online from any Internet-connected computer, or by a computer via download. With dynamic features built into this e-text, students can search across content, highlight, and take notes that they can share with teachers and classmates. Readers will also have access to interactive images and embedded podcasts.

Acknowledgments

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Thanks also to our families and friends for their patience and support along the way, especially to Christopher, Haley, and Hannah Wixom; and Alec Dennis; and Richard Jones, in loving memory.

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